

# PRINCIPLES OF PROGG LANG CS F301

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## Last Year's Tutorial for OO Language Comparisons: C/C++ #include v/s Java import demonstration

The tutorials, assignments and quizzes so far have demonstrated to you many program examples that reveal usually ignored aspects of programming languages and the corresponding paradigms. In this assignment, you are expected to start making such examples yourselves. Consider the questions carefully, and indeed look up the textbook, reference books and other authoritative sources (and some of the less reliable ones like LLM-based chatbots too) for answers. But as the assignment submission, I want you to give those example programs that you find in those sources or develop (and test) yourselves. With each such set of programs that you give in response to a question, include inline comments or some clear and complete, self-contained documentation that explains how the programs demonstrate the answer.

Questions for the "C/C++ Hash include v/s Java import demonstration" assignment.

1. Find an example of two headers (C/C++ .h files) in the libc/libc++ implementation on any platform (your laptop or any other) that shows the same type declared differently, with one declaration consistent with the type's implementation in the binary library and the other consistent with but restricted to the API available to you (as programmers). If there are no such headers (or there cannot be) in your opinion and search, argue why.
2. Answer for C, C++ and Java each: An array "a" is declared as "int a[11][3];". When all its elements are 0, and then the statement "a[12][1] = 1;" is encountered at that time, what will be the outcome? The possibilities are:
  - (a) a compilation error, or
  - (b) a runtime crash--no exception catch, or
  - (c) a runtime exception, or
  - (d) some "n"th element becomes 1.

In cases your answer is (d), also give that "n" each time. Each answer must be justified.

3. Find examples of a class in the Java sdk on your platform that has two different implementations (or dissimilar .class files) in two different packages.
4. Can each implementation of such a class be instantiated into two distinct objects inside the same method and block?
5. Is it possible to do so (3,4 above) in C++ as well? How? Without namespaces?

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## Submission status

|                          |                                   |
|--------------------------|-----------------------------------|
| <b>Attempt number</b>    | This is attempt 1.                |
| <b>Submission status</b> | No submissions have been made yet |
| <b>Grading status</b>    | Not graded                        |

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